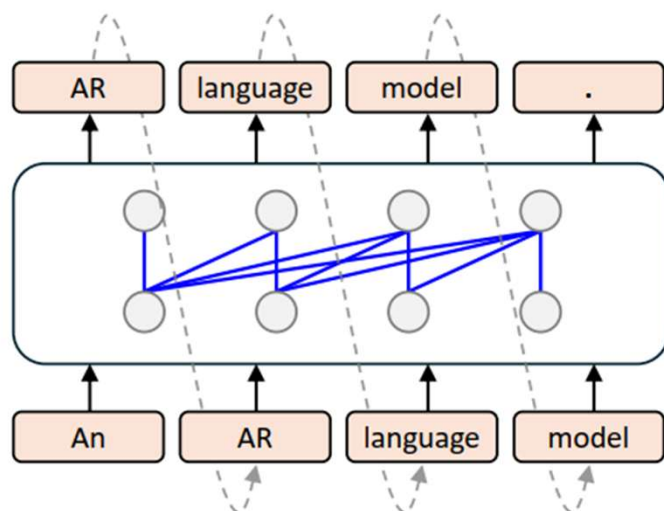


Diffusion vs. Autoregressive Language Models: A Text Embedding Perspective

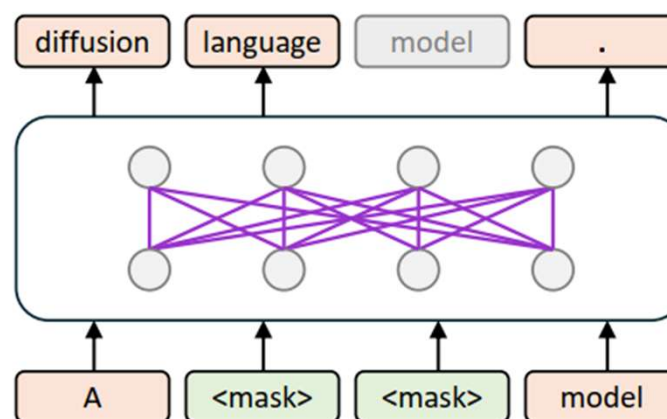
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¹Nanyang Technological University, ²Yale University,
³NYU Shanghai, ⁴Alibaba-NTU Singapore JRI

Introduction



(a) Autoregressive Modelling

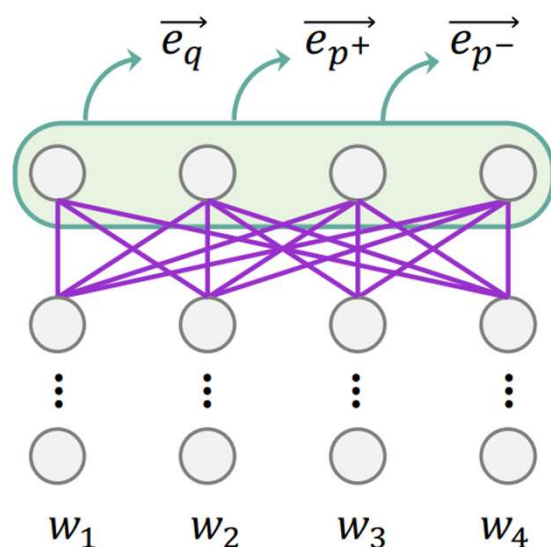


(b) Diffusion Modelling in DREAM

Our hypothesis: diffusion embeddings, when trained at scale with a bidirectional attention architecture, are better suited to **capturing global context** and thus achieve superior performance on **long and complex** documents.

Diffusion Embedding Model

- *DiffEmbed* extracts contextualized token representations from the backbone diffusion LM and aggregate them using mean pooling.



- The main difference between *DiffEmbed* and LLM-based embedding models is how they learn representations:
 - **LLM embeddings** are trained with causal attention and a next-token prediction task, so they mainly capture context from past to future.
 - **Diffusion LM embeddings** are trained with a denoising objective, learning to recover noisy or masked inputs all at once, which helps them capture bidirectional semantics and more robust representations.

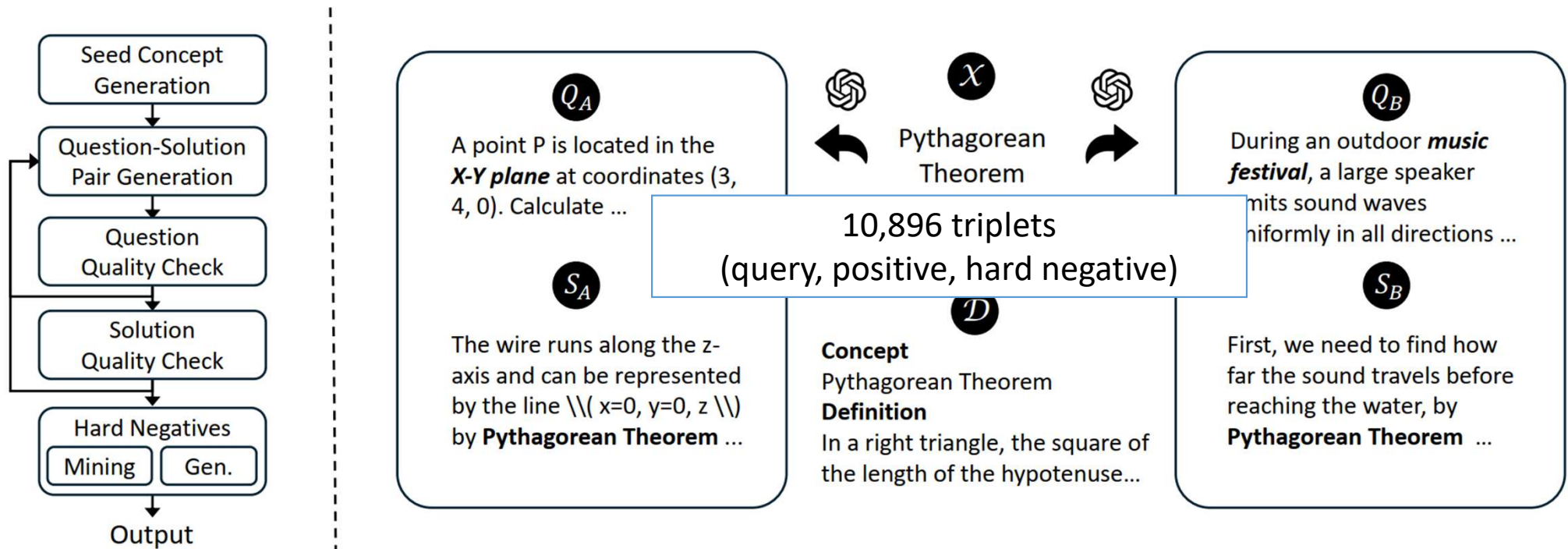
Evaluating Text Embedding Models (1)

- We compare *DiffEmbed* with LLM embedding models for diverse capabilities:

Task	Train Dataset	Test Dataset
Long-document Retrieval	E5 Public (Short & Long Docs)	LongEmbed
Reasoning-intensive Retrieval	<i>ReasonAug</i> (Ours, Medium Docs)	BRIGHT
Instruction-following Retrieval	MS MARCO w/ Instructions (Short Docs)	FollowIR
General Text Embedding Tasks	E5 Public (Short & Long Docs)	MTEB

Evaluating Text Embedding Models (2)

- Reasoning-intensive retrieval tasks the model to find relevant theorems given a question. However, there is no existing training set effective for this task at that time.



Experiment Setup

- **DiffEmbed**

We take DREAM-v0-Instruct-7B as the backbone model for *DiffEmbed*. Initialized from Qwen2.5-7B, DREAM is pre-trained on 580 billion tokens and achieves performance competitive with leading autoregressive LMs.

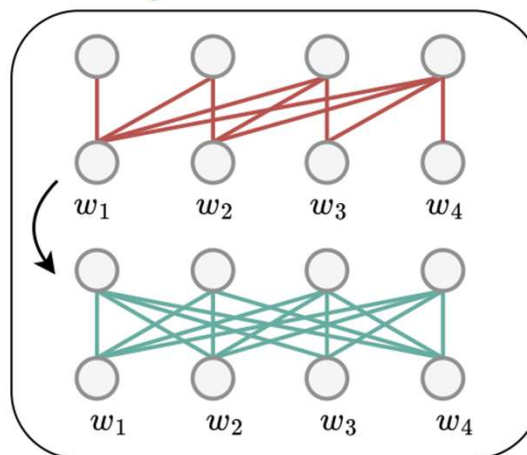
- Similarly sized Autoregressive LLMs

Llama-3-8B-Instruct
w/ LLM2Vec Adaptation

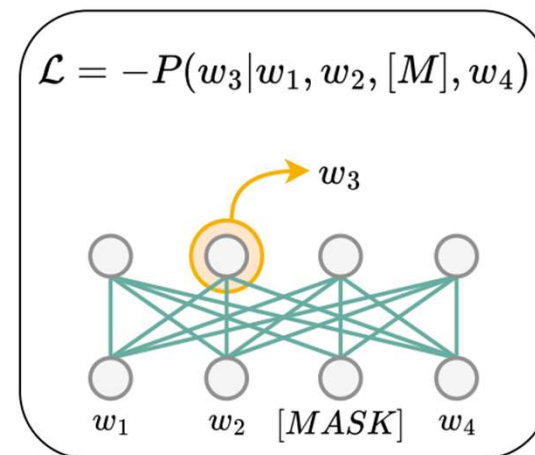
Mistral-7B-Instruct-v0.2
w/ LLM2Vec Adaptation

Qwen2.5-7B-Instruct
w/ LLM2Vec Adaptation

Enabling Bidirectional Attention



Masked Next Token Prediction



Results: Long Document Retrieval (1)

Train Dataset	Test Dataset
E5 Public	LongEmbed

Needle Test Examples:

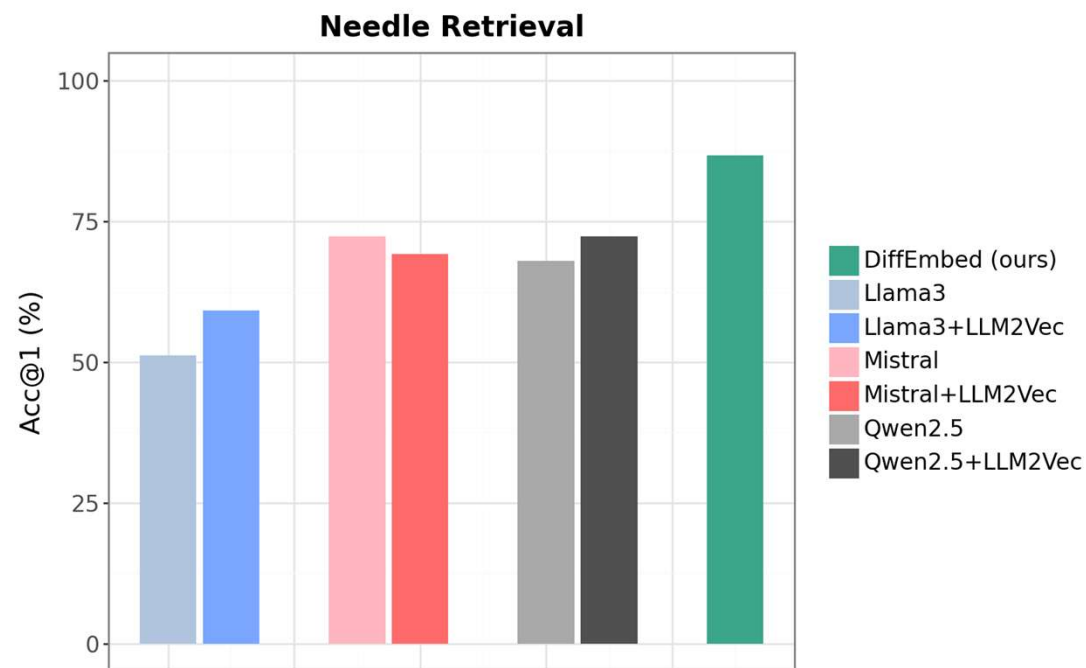
Query: Who discovered the law of gravity?

Doc1: <prefix> The law of gravity was discovered by Sir Issac Newton. <suffix>

Doc2: <prefix> The best thing to do in San Francisco is eat a sandwich and sit in Dolores Park on a sunny day. <suffix>

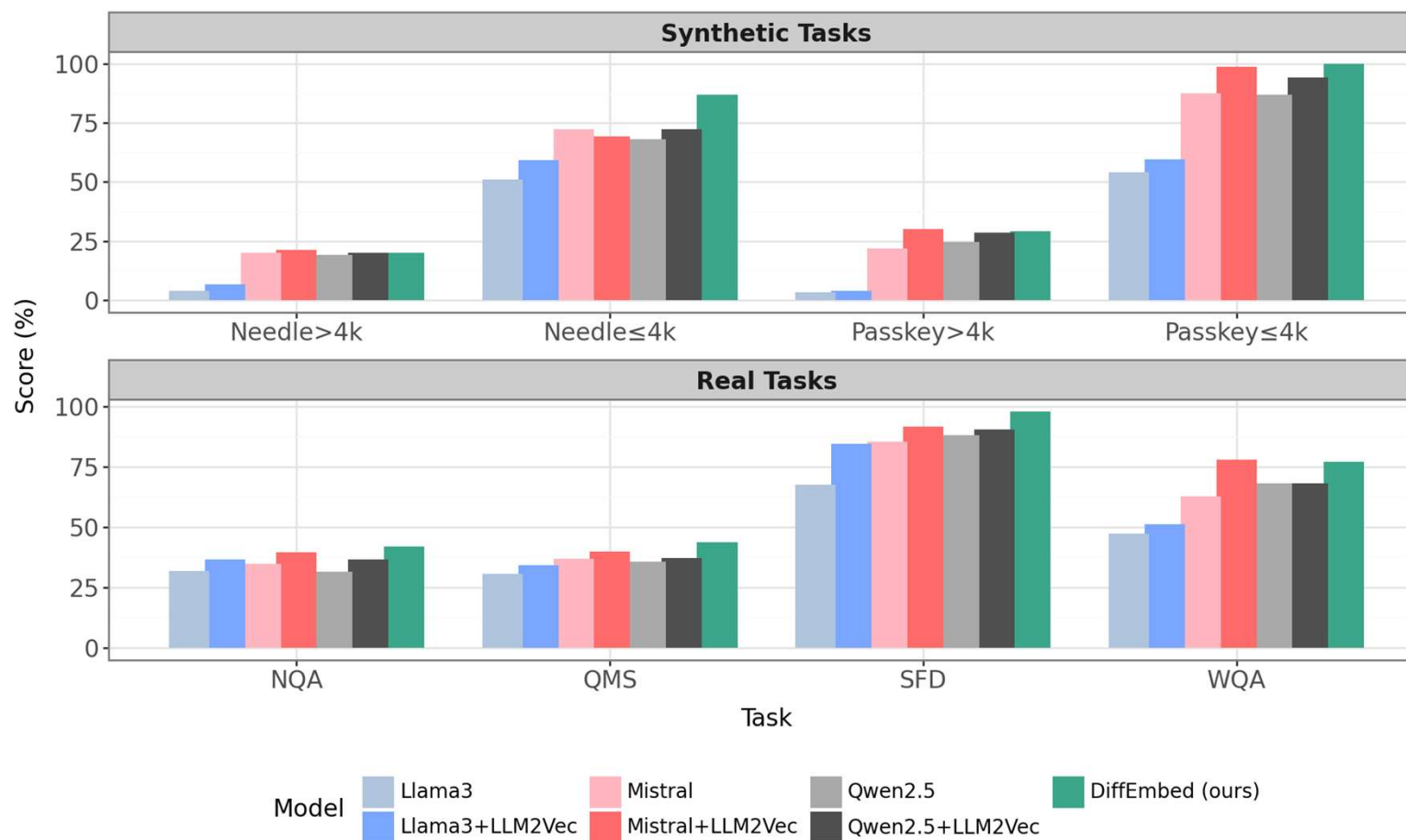
...

- LLM2Vec improves LLM embeddings by enabling bidirectional attention, which better captures long-range dependencies.
- **DiffEmbed** excels on long document retrieval including synthetic and real-world tasks without the need of adaptation.



Results: Long Document Retrieval (2)

Train Dataset	Test Dataset
E5 Public	LongEmbed



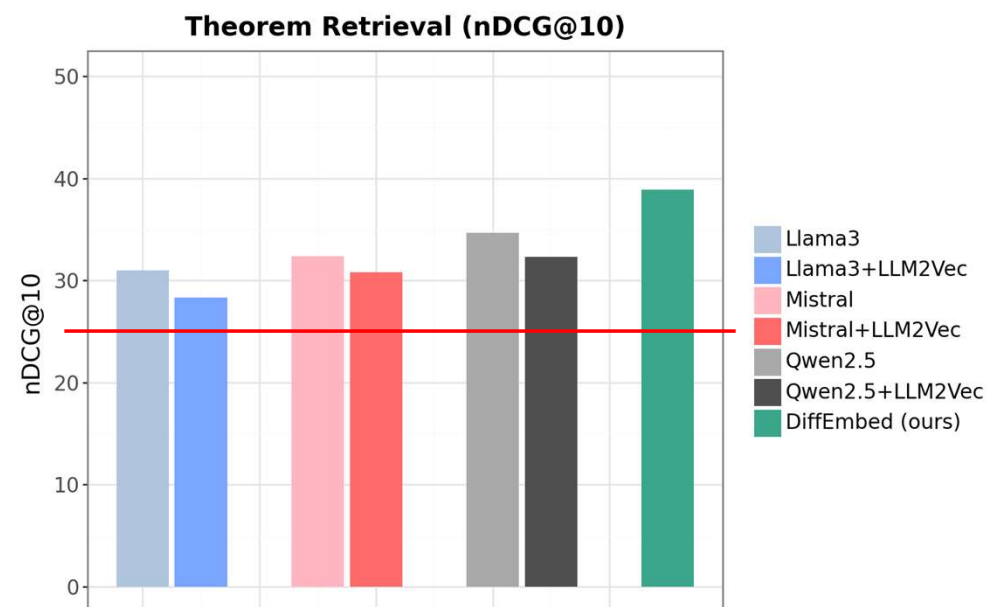
Results: Reasoning-Intensive Retrieval (1)

MATH - question

Determine all positive integers relatively prime to all the terms of the infinite sequence $a_n = 2^n + 3^n + 6^n - 1$...

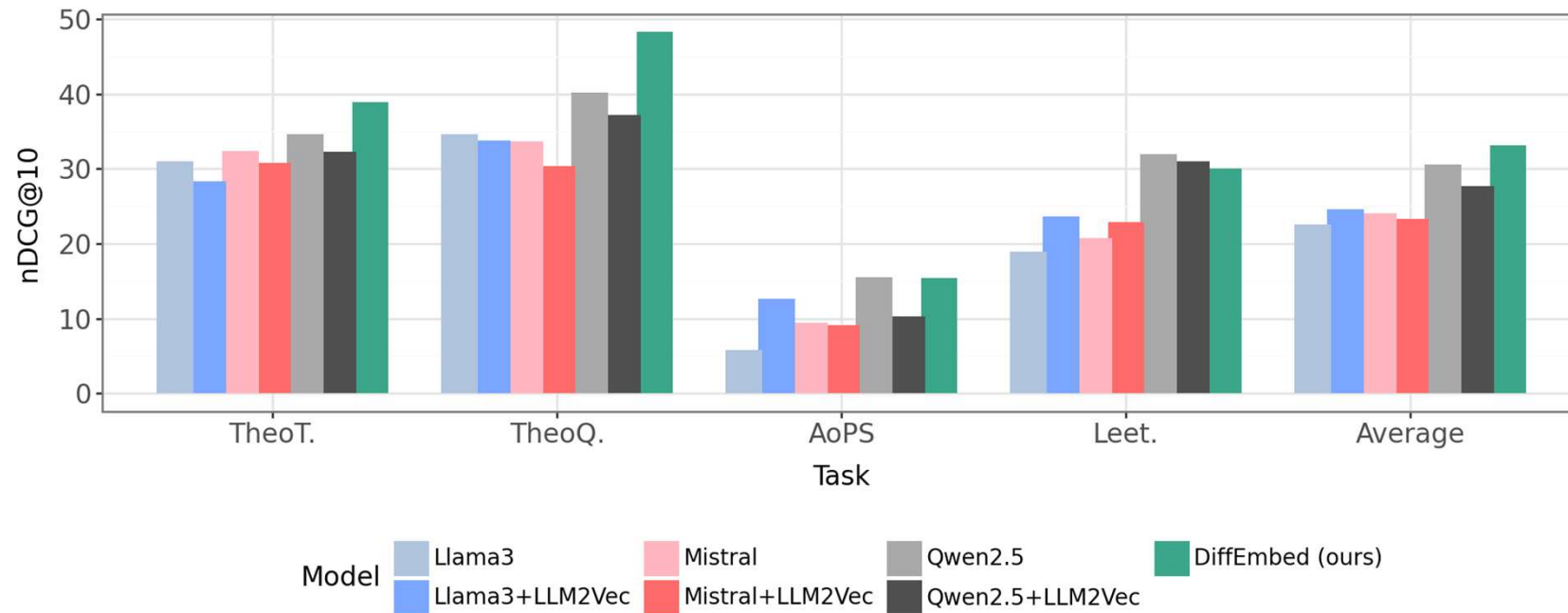
- LLM2Vec is ineffective, likely due to limited data scale and complexity.
- All fine-tuned embedding models outperform the general-purpose E5-Mistral (red line).
- **DiffEmbed** surpasses its base model Qwen2.5, highlighting that attending to both past and future context is crucial for understanding complex logic and equations.

Train Dataset	Test Dataset
ReasonAug	BRIGHT



Results: Reasoning-Intensive Retrieval (2)

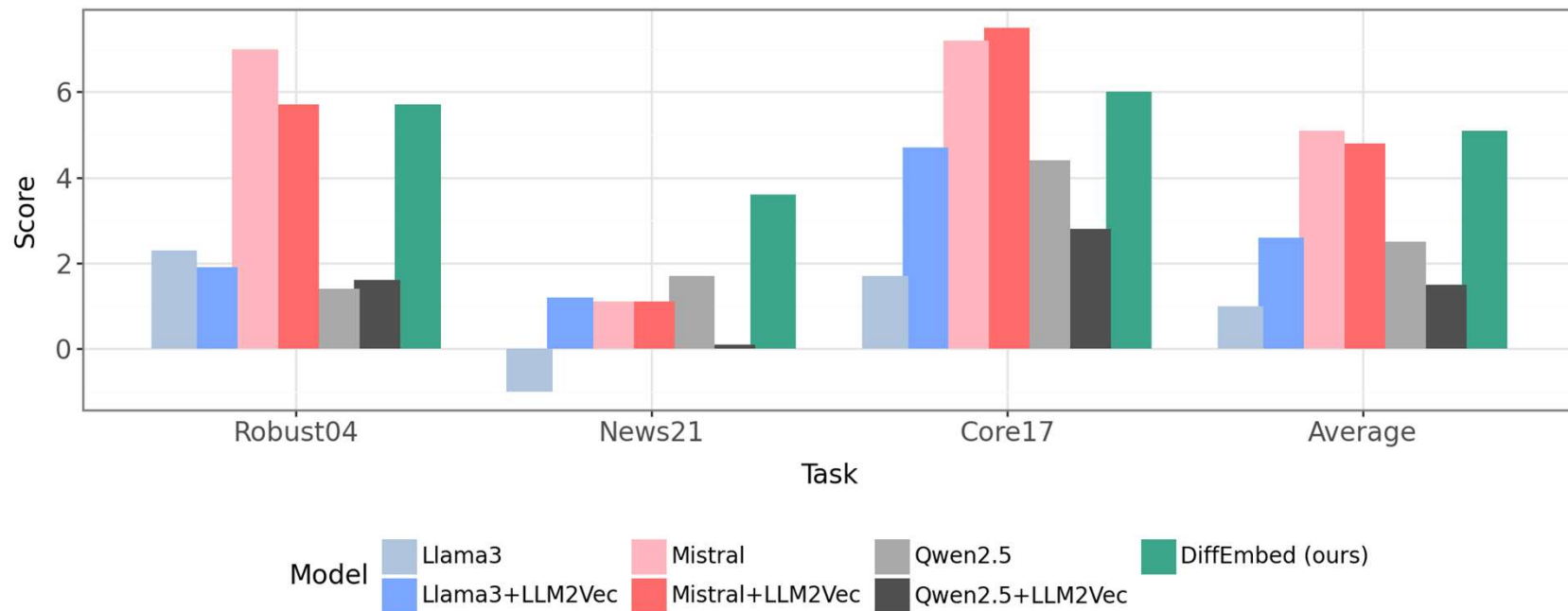
Train Dataset	Test Dataset
ReasonAug	BRIGHT



Results: Instruction-Following Retrieval

Train Dataset	Test Dataset
MS MARCO Instruct	FollowIR

- **DiffEmbed** matches the top-performing Mistral, with limited gains likely due to the short and simple queries and documents.



Results: General Text Embedding Tasks

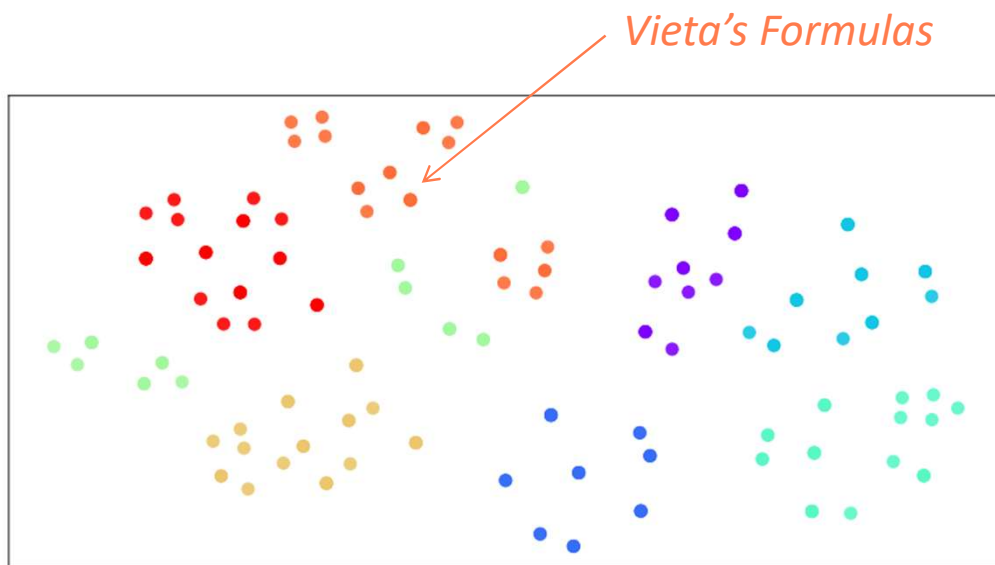
Train Dataset	Test Dataset
E5 Public	MTEB

- **DiffEmbed** performs comparably to LLM embedding models, as most conventional tasks have short inputs and limited reasoning, reducing its advantages.

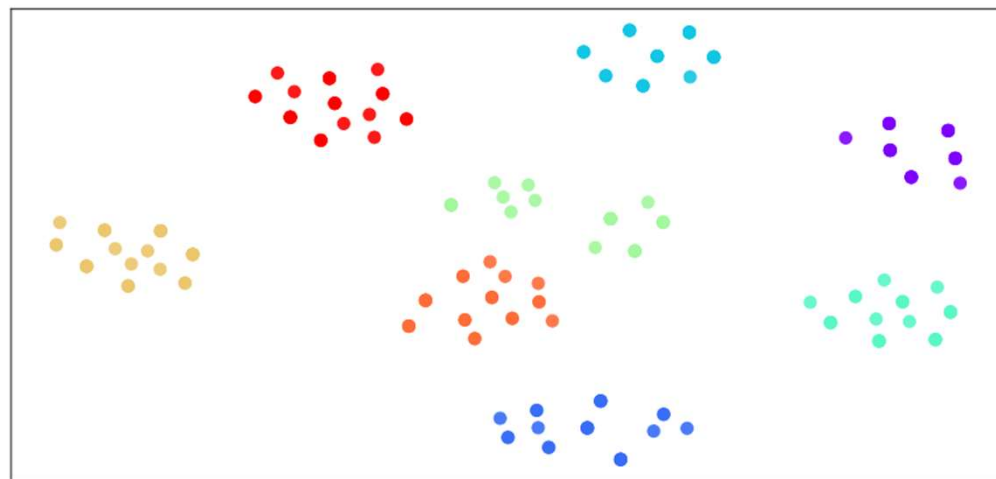
Task	Llama3	Mistral	Dream
ArguAna	57.33	54.46	58.47
Banking77Classification	87.35	87.03	87.50
BiorxivClusteringS2S	38.45	35.80	35.56
EmotionClassification	47.81	50.81	51.44
MassiveIntentClassification	78.69	76.89	75.64
MedrxivClusteringS2S	33.46	31.26	31.42
NFCorpus	38.04	39.27	35.44
SciDocsRR	85.19	84.41	82.00
SciFact	75.93	71.21	71.42
SICK-R	80.64	78.92	76.49
SprintDuplicateQuestions	92.30	95.28	95.30
StackOverflowDupQuestions	55.32	54.32	50.69
STS17	87.30	88.15	87.83
STSBenchmark	84.76	85.47	83.43
TwentyNewsgroupsClustering	53.28	49.50	52.56
Average	66.39	65.52	65.01

Q1: Why do generalist embedding models underperform in reasoning-intensive retrieval?

- When searching relevant questions, E5-Mistral often relies on superficial lexical cues (e.g., keywords like “smallest”) and shallow semantic patterns (e.g., “what is the count” vs. “how many”).



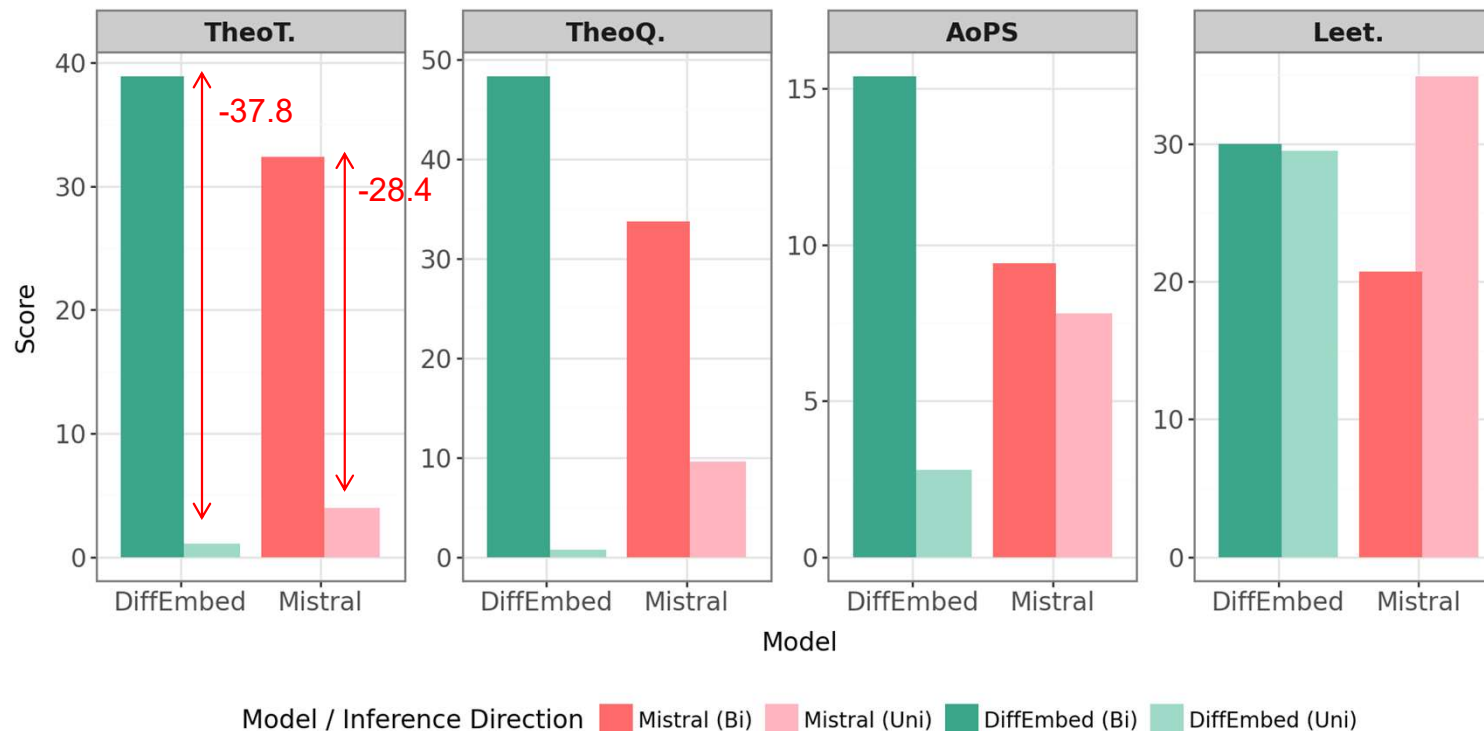
(a) E5-Mistral without REASONAUG fine-tuning.



(b) DIFFEMBED fine-tuned on REASONAUG.

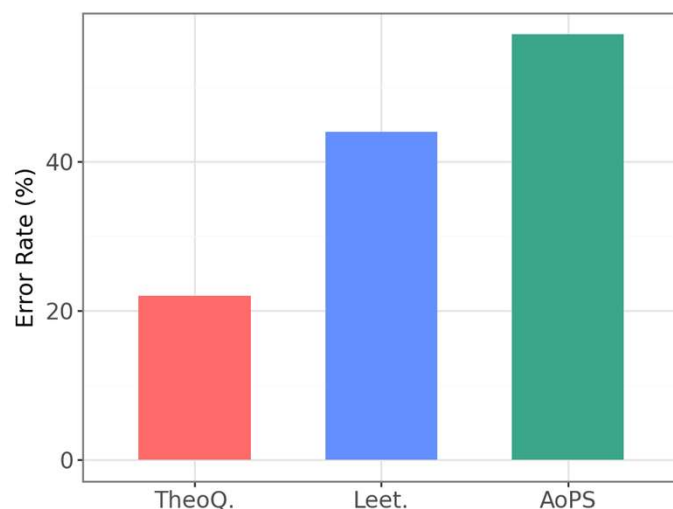
Q2: How important is bidirectional attention for different models and tasks?

- **DiffEmbed** exhibits a larger performance drop when reverse attention is disabled, which suggests that it relies more heavily on bidirectional context, likely due to its bidirectional pre-training.



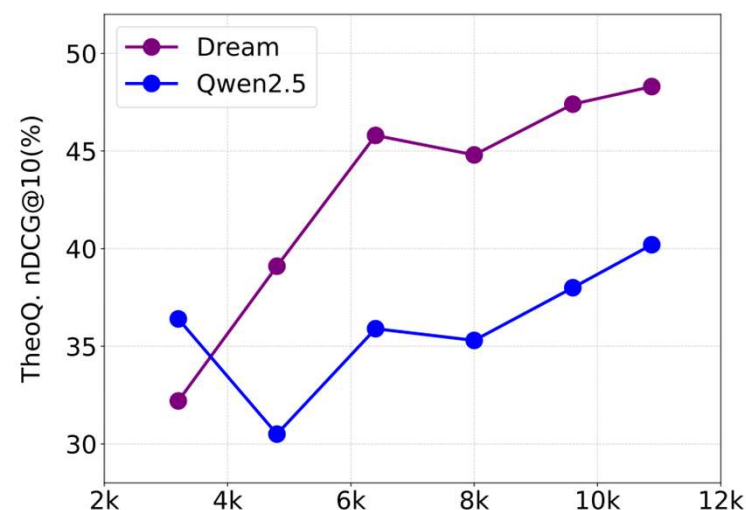
Q3: Why are improvements less evident in Leetcode and AoPS compared to TheoQ. In BRIGHT?

- Our analysis reveals notable noise in the gold annotations and corpora. Particularly we observe multiple types of relevance (*e.g.*, contextual and algorithmic) in Leet., while it is more consistent in TheoT. and TheoQ..



Q4: Does the performance gain diminish as the training dataset size increases?

- The performance gap remains substantial within this range, suggesting the LLM based models may need extensive data to learn effective bidirectional attention as *DiffEmbed*.



Diffusion vs. Autoregressive Language Models: A Text Embedding Perspective

Thank You!





- [1] BehnamGhader P, Adlakha V, Mosbach M, et al. Llm2vec: Large language models are secretly powerful text encoders[J]. arXiv preprint arXiv:2404.05961, 2024.
- [2] Nie S, Zhu F, You Z, et al. Large language diffusion models[J]. arXiv preprint arXiv:2502.09992, 2025.
- [3] Achiam J, Adler S, Agarwal S, et al. Gpt-4 technical report[J]. arXiv preprint arXiv:2303.08774, 2023.



Results: Long Document Retrieval

Train Dataset	Test Dataset
E5 Public	LongEmbed

	N	Synthetic (Acc@1)				Real (nDCG@10)				Avg.
		$Pk \leq 4k$	$Pk > 4k$	$Nd \leq 4k$	$Nd > 4k$	NQA	QMS	SFD	WQA	
E5-Mistral	2M	95.6	30.0	68.8	14.0	44.6	43.6	96.8	82.0	59.4
Llama3	16k	54.0	3.3	51.2	4.0	31.9	30.8	67.6	47.3	32.3
+ LLM2Vec	16k	59.6	4.0	59.2	6.7	36.5	34.4	84.5	51.1	42.0
Mistral	16k	87.6	22.0	72.4	20.0	35.0	36.8	85.4	62.9	52.8
+ LLM2Vec	16k	98.8	30.0	69.2	21.3	39.5	40.0	91.6	78.0	58.6
Qwen2.5	16k	86.8	24.6	68.0	19.3	31.7	35.6	88.0	68.3	52.8
+ LLM2Vec	16k	94.0	28.7	72.4	20.0	36.5	37.3	90.5	68.1	55.9
DIFFEMBED	16k	100	29.3	86.8	20.0	42.1	43.8	98.0	77.2	62.2

Table 2: Results on long document retrieval (LONGEMBED) for embedding models trained with **E5** text embedding data. N is the amount of E5 samples used for training. $Pk \leq 4k$ refers to Passkey Retrieval with an evaluation length less than or equal to 4,096. $Nd > 4k$ refers to Needle-in-a-haystack Retrieval with length larger than 4,096. NQA , QMS , SFD , WQA is short for *NarrativeQA*, *QMSum*, *SummScreenFD*, *2WikiMultihopQA*, respectively. The best results with 16k training data are in **bold**.

Results: Reasoning-Intensive Retrieval (1)

Train Dataset	Test Dataset
ReasonAug	BRIGHT

	N	TheoT.	TheoQ.	AoPS	Leet.	Average
E5-Mistral	-	26.8	26.1	7.1	28.7	22.2
ReasonIR	-	27.2	31.9	14.7	35.0	27.2
Llama3	10k	31.0	34.6	5.8	18.9	22.6
+ LLM2Vec	10k	28.3	33.8	12.6	23.6	24.6
Mistral	10k	32.4	33.7	9.4	20.7	24.1
+ LLM2Vec	10k	30.8	30.4	9.1	22.9	23.3
Qwen2.5	10k	34.7	40.2	15.5	32.0	30.6
+ LLM2Vec	10k	32.3	37.2	10.3	31.0	27.7
DIFFEMBED	10k	38.9	48.3	15.4	30.0	33.2

Table 3: Results on theorem-based benchmarks in reasoning-intensive retrieval (BRIGHT) for embedding models trained using REASONAUG. We report nDCG@10 for *TheoremQA* with theorem retrieval (*TheoT.*) and question retrieval (*TheoQ.*), *AoPS*, and *LeetCode* (*Leet.*). N is the amount of REASONAUG samples used for training. The best results with 10k training data are in **bold**.

Results: Instruction-Following Retrieval

Train Dataset	Test Dataset
MS MARCO Instruct	FollowIR

	N	Robust04		News21		Core17		Average	
		MAP	MRR	nDCG	MRR	MAP	MRR	Score	MRR
Prompt-riever	1M	28.3	+11.7	28.5	+6.4	21.6	+15.4	26.1	+11.2
Llama3	16k	12.7	+2.3	17.6	-1.0	11.7	+1.7	14.0	+1.0
+ LLM2Vec	16k	17.6	+1.9	21.6	+1.2	16.7	+4.7	18.6	+2.6
Mistral	16k	21.4	+7.0	27.4	+1.1	16.0	+7.2	21.5	+5.1
+ LLM2Vec	16k	20.7	+5.7	25.1	+1.1	16.3	+7.5	20.7	+4.8
Qwen2.5	16k	15.6	+1.4	20.4	+1.7	13.9	+4.4	16.6	+2.5
+ LLM2Vec	16k	14.5	+1.6	19.7	+0.1	13.8	+2.8	16.0	+1.5
DIFFEMBED	16k	18.9	+5.7	27.7	+3.6	16.2	+6.0	20.9	+5.1

Table 4: Results on instruction-following document retrieval (FOLLOWIR) for embedding models trained using **MS MARCO with Instructions**. N is the amount of MS MARCO samples used for training. MAP@1000/nDCG@5 range from 0-100. MRR is short for the pairwise evaluation metric measuring instruction following when instructions change, ranging from -100 to 100 (higher is better). The best results trained with 16k data are in **bold**.

Q2: How important is bidirectional attention for different models and tasks?

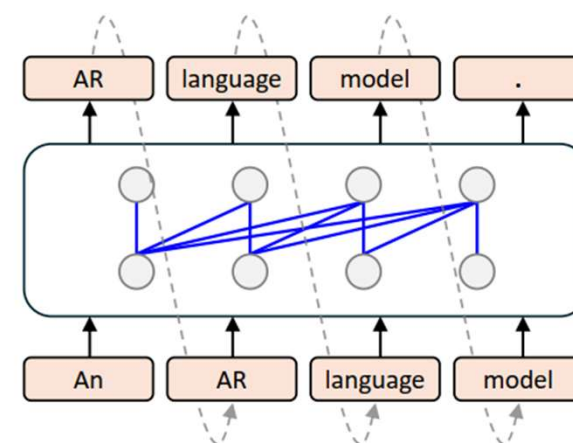
- We ablate two embedding models, Mistral and DIFFEMBED on BRIGHT. For each model, we compare testing performance using full bidirectional (causal and reverse) attention versus unidirectional (causal only) attention.
- DIFFEMBED exhibits a substantially **larger performance drop when reverse attention is disabled**, which suggests that DIFFEMBED depends more heavily on bidirectional context, likely due to its bidirectional pre-training.
- Tasks differ in their sensitivity to reverse attention. Performance on the Leetcode task remains relatively stable (even increase) without reverse attention, while performance on the TheoremQuestions task degrades significantly.

	Bidirectional	Unidirectional
Mistral		
TheoT.	32.4	4.0 (-28.4)
TheoQ.	33.7	9.6 (-24.1)
AoPS	9.4	7.8 (-1.6)
Leet.	20.7	34.9 (+14.2)
DIFFEMBED		
TheoT.	38.9	1.1 (-37.8)
TheoQ.	48.3	0.7 (-47.6)
AoPS	15.4	2.8 (-12.6)
Leet.	30.0	29.5 (-0.5)

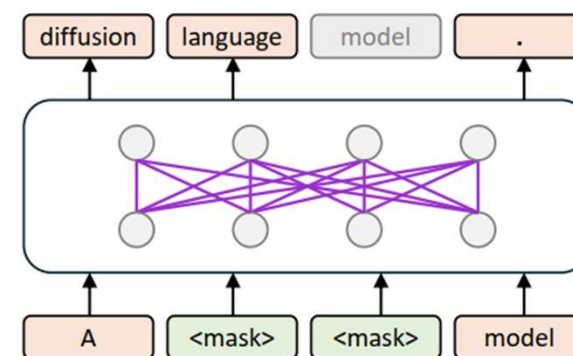
Table 5: Ablation study comparing BRIGHT performance when using bidirectional vs. unidirectional attention at test time. Both models are trained with bidirectional attention and REASONAUG data.

Introduction

- LLM-based embedding models, benefiting from large scale pretraining and post-training, have begun to surpass BERT and T5-based models on general purpose text embedding tasks such as document retrieval.
- A fundamental limitation lies in the unidirectional attention used during **autoregressive** pre-training, which misaligns with the bidirectional nature of text embedding tasks. Prior works adapt LLMs with bidirectional attention through continue training with a limited scale [1].
- We propose adopting diffusion language models for text embeddings, motivated by their **inherent bidirectional** architecture and recent success in matching or surpassing LLMs especially on reasoning tasks [2].
- *Our hypothesis: diffusion embeddings, when trained at scale with a bidirectional attention architecture, are better suited to **capturing global context** and thus achieve superior performance on **long and complex documents**.*



(a) Autoregressive Modelling



(b) Diffusion Modelling in DREAM

- [1] BehnamGhader P, Adlakha V, Mosbach M, et al. Llm2vec: Large language models are secretly powerful text encoders[J]. arXiv preprint arXiv:2404.05961, 2024.
- [2] Nie S, Zhu F, You Z, et al. Large language diffusion models[J]. arXiv preprint arXiv:2502.09992, 2025.

Results: Reasoning-Intensive Retrieval (3)

Train Dataset	Test Dataset
ReasonIR	BRIGHT

	Bio.	Ear.	Eco.	Psy.	Rob.	Sta.	Sus.	Leet.	Pony	AoPS	TQ.	TT.	Avg.
E5-Mistral	18.8	26.0	15.5	15.8	16.4	9.8	18.5	28.7	4.8	7.1	26.1	26.8	17.9
ReasonIR	26.2	31.4	23.3	30.0	18.0	23.9	20.5	35.0	10.5	14.7	31.9	27.2	24.4
Llama3	7.5	28.2	18.1	24.5	16.1	21.7	17.0	37.0	4.3	6.3	26.2	21.7	19.1
Qwen2.5	12.5	32.1	19.6	20.1	14.2	19.4	13.7	40.6	1.8	10.1	36.5	28.8	20.8
DIFFEMBED	14.7	31.5	19.6	25.9	16.1	20.2	17.2	41.3	2.5	8.4	39.6	32.6	22.5

Table 10: Results on reasoning-intensive retrieval (BRIGHT) for embedding models trained using ReasonIR data. We report nDCG@10 for each subset task. We fine-tune the LLMs using LoRA instead of full parameter updates like in ReasonIR, leveraging 100k samples each from Public E5, ReasonIR VL, and ReasonIR HQ datasets.